Training Day 12

1. Topics Covered and Brief Explanation

1. Introduction to Linear Regression

Linear Regression is a supervised machine learning algorithm used to predict continuous numerical values based on the relationship between input and output variables.

For example, predicting house prices based on their area or estimating student marks based on study hours.

2. Dependent and Independent Variables

- Independent Variable (X): The input variable used to make predictions.
- Dependent Variable (Y): The output variable that needs to be predicted.

For example, when predicting marks based on study hours, study hours represent X and marks represent Y.

3. Linear Regression Equation

Studied the basic mathematical equation of Linear Regression:

\[Y = mX + c \]

Where:

- Y = Predicted output.
- X = Input variable.
- m = Slope of the line.
- c = Intercept.

4. Best-Fit Line

Learned how Linear Regression finds the best-fit straight line that represents the relationship between input and output data. The model attempts to minimize the difference between actual and predicted values.

5. Applications of Linear Regression

- House price prediction.
- Sales forecasting.
- Student performance prediction.
- Temperature estimation.
- Business trend analysis.

2. Tools and Technologies Used

Tool	Purpose
Python	Implementing regression programs.
Scikit-learn	Creating and training Linear Regression models.
NumPy	Handling numerical data.
Matplotlib	Visualizing regression results.
Google Colab	Writing and executing Python code.

3. Practical Learning / Code Example

Example: Predicting Marks Based on Study Hours

```
from sklearn.linear_model import LinearRegression
import matplotlib.pyplot as plt

# Input data
X = [[1], [2], [3], [4], [5]]
# Output data
y = [20, 40, 60, 80, 100]
# Create and train model
model = LinearRegression()
model.fit(X, y)
# Predict marks
prediction = model.predict([[6]]
print("Predicted Marks:", prediction[0])
# Visualize data
plt.scatter(X, y)
plt.plot(X, model.predict(X))
plt.xlabel("Study Hours")
plt.ylabel("Marks")
plt.title("Linear Regression")
plt.show()
```

Explanation:

- LinearRegression() creates a regression model.
- fit() trains the model using the provided data.
- predict() estimates marks for six hours of study.
- scatter() displays the actual data points.
- plot() displays the fitted regression line.

4. Learning Outcomes

- Understood the concept and importance of Linear Regression.
- Learned about dependent and independent variables.
- Understood the basic equation of a regression model.
- Gained knowledge of how a best-fit line is generated.
- Practised implementing and visualizing a Linear Regression model using Scikit-learn and Matplotlib.